OSSP HACKATHON 2026 — TECHNICAL DOCUMENTATION

TLB and Page Table Architecture System Manual

Setup Instructions • Architecture Specification • Test Verification

1. System Overview & Dual-Environment Deployment

The simulator is packaged as a dual-environment tool designed for maximum portability and demonstration flexibility:

• **Web Visual Dashboard (index.html):** Built in Vanilla HTML5, CSS3, and JavaScript. Runs instantly in any web browser with zero installation, external servers, or package dependencies.
• **Terminal CLI Engine (simulator.py):** Built with standard Python 3.6+. Provides an interactive REPL command prompt, formatted ASCII memory table dumps, and automated trace verification.

2. Execution Guide

Environment	Launch Command	Purpose
Web Visualizer	`Start-Process index.html`	Full interactive GUI with pipeline animations & live tables
Slide Presentation	`Start-Process presentation.html`	Interactive in-browser slide deck for evaluator presentation
Python CLI (Default)	`py simulator.py`	Interactive terminal shell with ASCII tables & prompt
Python CLI (Custom)	`py simulator.py --pagesize 1024 --tlb 8`	Custom architectural parameter configuration

3. Hardware Specifications & Configurable Parameters

Hardware Parameter	Default Value	Permitted Range	Architectural Description
Virtual Address Space	16-bit (64 KB)	Fixed (16-bit)	Addresses from 0x0000 to 0xFFFF
Page Size (P)	4096 Bytes	256B, 1KB, 4KB	Determines offset bits k = log2(P)
TLB Cache Capacity	4 Slots	2 to 16 Slots	Fully associative parallel tag search
Replacement Policy	LRU	LRU, FIFO	Cache eviction on capacity saturation
Physical RAM Frames	8 Frames (32 KB)	4 to 16 Frames	Total allocatable physical memory frames
TLB Latency (t_TLB)	2 ns	1 to 100 ns	SRAM on-chip cache search time
RAM Latency (t_MEM)	100 ns	10 to 1000 ns	Main DRAM read/write cycle time
Page Fault Penalty	8000 ns	1000 to 50M ns	OS trap handler and secondary storage I/O

4. Step-by-Step Live Demo Test Cases

Test Scenario	Input VA	VPN / Offset	Expected Hardware Behavior & Result

Test 1: TLB Hit	`0x1A4C`	VPN: 1 Offset: 0xA4C	**TLB HIT** in Slot #0. Maps to Frame #2. Physical Address = **0x2A4C**. Access Time = 102 ns.
Test 2: Page Table Hit	`0x0080`	VPN: 0 Offset: 0x080	**TLB MISS** ➔ Page Table Hit in Frame #0. TLB Slot #1 updated. PA = **0x0080**. Latency = 202 ns.
Test 3: Page Fault	`0x5100`	VPN: 5 Offset: 0x100	**PAGE FAULT** (Page 5 not in RAM). Frame #3 allocated. TLB updated. PA = **0x3100**.
Test 4: Cache Eviction	`Batch Trace`	Pages 0..4 in loop	Demonstrates eviction of oldest/least used slot when 5th page is loaded into 4-slot TLB.